Appendix 1 – Features, Threshold Values and Number of Buildings

Feature	Category	Threshold	# Buildings
Height	Low	≤8	19,864
	Mid	≤18	17,345
	High	≤30	1,592
	Extreme	>30	190
Floorspace	Low	≤100	7,921
	Mid	≤400	16,010
	High	≤1800	12,852
	Extreme	>1800	2,208
Building salience	Low	>0.3	33,197
	High	≤0.3	5,794
Distance to city center	Low	≤2500	10,897
	Mid	≤5000	18,685
	High	≤7000	8,268
	Extreme	>7000	1,141
Distance to nearest subcenter	Very low	≤1000	11,820
	Low	≤1500	8,777

Information Classification: General

	Mid	≤2250	8,809
	High	3100	5,730
	Extreme	>3100	3,855
Household income	No Data	No households in vicinity	2,959
	Very low	≤7000	12,589
	Low	≤13000	11,507
	Medium	≤22500	7,240
	High	≤35000	2,969
	Extreme	>35000	1,457
Percent of children	No Data	No households in vicinity	2,959
	Low	≤0.353	313
	Medium	≤0.5954	15,015
	High	>0.5954	20,704
Percent of senior	No Data	No households in vicinity	2,959
	Low	≤0.4	35,961
	High	>0.4	71
Degree	No Data	Nearest junction is more than 250m away	250
	Low	<2	26

Information Classification: General

	Medium	<6	22,532
	High	>=6	16,183
Betweenness centrality	No Data	Nearest junction is more than 250m away	250
	Low	<232,124	33,536
	Medium	<856,008	4,288
	High	<4,149,076	868
	Extreme	>=4,149,076	49
Distance to region 1	Low	0	10,697
	Medium	<3,522	14,890
	High	>=3,522	13,404
Distance to region 2	Low	0	8,027
	Medium	<1806	13,237
	High	<3793	11,216
	Extreme	>=3793	6,511
Distance to region 3	Low	0	12,284
	Medium-low	<954	6,753
	Medium	<1925	6,538
	Medium-high	<2967	8,157

Information Classification: General

	High	<4249	3,868
	Extreme	>=4249	1,391
Distance to region 4	Low	0	7,819
	Medium	<2389	11,995
	High	<5637	10,294
	Extreme	>=5637	8,883

Appendix 2 – examples of generated trajectories

<Figure A2: Generated trajectories of four randomly selected agents, starting from home and going back to home. The gradient of the colors (from light to dark) complements the annotated numbers which indicate the order of the edges. To prevent clutter, trajectories with fewer than three edges are not annotated>

Appendix 3 – City structure experiment: Reduced Boundaries

<Figure A3 Connectivity Plot- the top 0.5% connectivity of the remaining buildings in the study area with reduced boundaries>

Configuration	Model	Total Trajectory Distance (log)	Mean Trajectory Segment Distance (log)	Trajectory Length (Buildings)	Convex Hull Area (log)	Inverse CPC (Building Type)	NFM (RMSE)
2	Untrained	3.35	1.12	2.28	6.08	**0.45**	22.89
	Linear	3.26	1.03	2.28	5.91	0.46	22.54
	MLP	3.27	1.04	2.28	5.95	0.46	22.52
	GCN	**2.78**	**0.60**	2.28	**4.98**	0.50	22.75
	GraphConv	2.98	0.77	**2.27**	5.39	0.48	**22.27**
3	Untrained	2.16	1.96	**4.26**	3.64	0.37	17.52
	Linear	2.03	1.84	4.48	3.37	0.37	16.12
	MLP	2.04	1.86	4.37	3.39	**0.36**	17.29
	GCN	**1.64**	**1.44**	4.63	**2.64**	0.40	**15.59**
	GraphConv	1.78	1.59	4.63	2.88	0.39	16.73

Table A3. Metrics for generated trajectories, for the four reward models and an untrained transition matrix, under two terminal state configurations. Reduced boundaries experiment

Information Classification: General

Appendix 4: City structure experiment – Random Removal

< Figure A4: The top 0.5% connectivity of the remaining buildings in the study area with 30% random removal>

Configuration	Model	Total Trajectory Distance (log)	Mean Trajectory Segment Distance (log)	Trajectory Length (Buildings)	Convex Hull Area (log)	Inverse CPC (Building Type)	NFM (RMSE)
2	Untrained	1.68	**0.61**	2.28	2.82	0.44	23.06
	Linear	1.67	0.62	2.28	2.78	0.44	22.72
	MLP	1.67	0.62	**2.28**	2.80	**0.43**	22.47
	GCN	**1.58**	0.70	2.28	**2.60**	0.46	**22.27**
	GraphConv	1.67	0.62	2.28	2.78	0.44	22.48
3	Untrained	0.52	0.34	**4.58**	1.23	0.37	18.49
	Linear	0.50	0.32	4.64	1.20	0.37	**16.32**
	MLP	0.48	0.31	4.67	1.17	**0.36**	16.43
	GCN	**0.39**	**0.22**	4.71	**1.08**	0.38	24.12
	GraphConv	0.45	0.27	4.79	1.14	0.37	18.24

Table A4. Metrics for generated trajectories, for the four reward models and an untrained transition matrix, under two terminal state configurations. Random removal experiment.

Appendix 5: City structure experiment – Building Editing

< Figure A5.1: The top 0.5% connectivity of buildings in the study area after building editing>

< Figure A5.2. Normalized floorspace vs. normalized final rewards for (a) linear model, (b) MLP, (c) GraphConv, (d) GCN, under the building editing experiment>

< Figure A5.3. Normalized original (no experiment) rewards vs. normalized final (experiment) rewards for (a) linear model, (b) MLP, (c) GraphConv, (d) GCN, under the building editing experiment>

Information Classification: General

Configuration	Model	Total Trajectory Distance (log)	Mean Trajectory Segment Distance (log)	Trajectory Length (Buildings)	Convex Hull Area (log)	Inverse CPC (Building Type)	NFM (RMSE)
2	Untrained	3.18	0.96	2.28	5.81	**0.45**	22.50
	Linear	3.08	0.86	2.28	5.60	0.46	**21.63**
	MLP	3.10	0.88	2.28	5.65	0.46	21.78
	GCN	**2.73**	**0.56**	**2.28**	**4.95**	0.49	21.97
	GraphConv	2.82	0.63	2.28	5.11	0.48	22.24
3	Untrained	1.95	1.75	**4.60**	3.23	**0.39**	**16.98**
	Linear	1.86	1.66	4.65	3.04	0.40	19.86
	MLP	1.81	1.61	4.81	2.92	0.40	20.76
	GCN	**1.45**	**1.25**	5.02	**2.24**	0.46	17.13
	GraphConv	1.55	1.35	4.90	2.42	0.43	21.11

Table A5. Metrics for generated trajectories, for the four reward models and an untrained transition matrix, under two terminal state configurations. Building editing experiment.

Appendix 6: Sensitivity Tests – Feature Matrix Definitions Model Parameters

<Figure A6.1. Rewards for four sensitivity tests vs. rewards for the original model. (a) reducing threshold for 'high' floorspace class to 1,000; (b) raising the thresholds for the 'mid' and 'high' floorspace classes to 500 and 1,000 respectively; (c) reducing all household income thresholds by 5,000NIS; (d) the 'low', 'medium', and 'high' income thresholds to 15,000NIS, 18,5000NIS, and 23,000NIS respectively>

<Figure A6.2. Rewards for two sensitivity tests vs. rewards for the original model. Upper row – more neurons, lower row – more layers. (a), (d) – MLP; (b), (e) – GCN; (c), (f) – GraphConv>

Information Classification: General